Lab 5

# Pathogenwatch and Microreact: Many Genomes at Once

Day 2 · 11:30–12:30 · 60 min
Pairs

LO5: use free web platforms to type, screen and visualise a collection of genomes

# Open your handout

## Lab 5 handout

`Lab5_pathogenwatch_microreact.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

# The scenario

- The IPC nurse at **District Hospital A** is worried about more **MRSA SSIs on Surgical Ward B**
- The lab stored the isolates and sent them for sequencing
- You have **genomes from all MRSA isolates of the past months**

> **KEY MESSAGE**
>
> **Get a quick overview of all the genomes at once.**

The genomes are real public data. The hospital, wards and dates are fictional.

# Plan for this lab

| Part | What | Time |
|---|---|---|
| A | Pathogenwatch: typing and AMR | 30 min |
| B | Microreact: tree + timeline | 25 min |
| | Summary | 5 min |

# Files from the instructor

| File | Contains |
|---|---|
| `assemblies/` | One FASTA per isolate |
| `koser_tree.nwk` | Phylogenetic tree (Galaxy: Snippy → IQ-TREE) |
| `lab6_metadata.csv` | Patient, ward, procedure and date information |

- **GitHub:** github.com/aryakaul/ughe-ssi-genomics-course
- Green **Code** button → **Download ZIP** → unzip → `labs/data/`

# Upload and explore

1. Log in to `pathogen.watch` → **Upload** → drag in the FASTA assemblies
2. Wait for each genome to be analysed (a few minutes)
3. For **at least 6** genomes, record:

| Genome | Species | ST | Predicted resistances | QC pass? |
|---|---|---|---|---|
| | | | | |
| | | | | |

Does the organism agree with the lab? Check the assembly QC metrics (length, contigs, N50).

# Make a collection and view the tree

1. Select all your genomes → **Create collection** → name it `UGHE Lab 5`
2. Open the collection → explore the **tree** view
3. Explore the **AMR** view: resistance profiles side by side

**CHECK YOURSELF**

1. How many different STs are in the collection?
2. Do all isolates have the **same** AMR profile? What does that suggest, and what does it **not** prove?
3. In the tree, can you see a group of isolates on very **short** branches?

# Create a project

1. Go to `microreact.org` → **Upload** (sign in only if you want to save)
2. Drag in **both** files: `lab6_metadata.csv` and `koser_tree.nwk`
3. Column that links the table to the tree tips → **`id`**
4. Dates: confirm that **`year`**, **`month`**, **`day`** form the date
5. **Continue / Create**

> “No matching IDs”? Raise your hand: the tree labels and the `id` column must match exactly.

# Explore: tree + time + place

- **Colour** the tree by `ward`, then `theatre`, then `hospital`
- Open the **Timeline**: when did each infection happen?
- **Click a tip**: its row lights up in the table and timeline
- **Lasso-select** a group of close tips: where and when are they?

**CHECK YOURSELF**

1. Is there a group that is closely related **and** close in time **and** in the same place?
2. Is any isolate from the **same ward at the same time not** in that group?
3. Is any isolate in the group from a **different ward**? What could link it?

PART B3

# Save, share, summarise

**Save** (if signed in) and copy the project link. **You'll use it in Lab 6.**

| Tool | Gives you | Input |
|---|---|---|
| **Pathogenwatch** | Species, ST, AMR, QC, clustering, global context | FASTA assemblies |
| **Microreact** | Visual story: tree + time + place + metadata | Tree (Newick) + table (CSV) |

Both are free and used by national public-health labs. **You can use them on Monday with your own data.**

Pathogenwatch: Centre for Genomic Pathogen Surveillance. Microreact: Argimón S, et al. Microb Genom 2016.